

Real-Time FPGA Implementation of a QPSK Modulator and Demodulator with Simulated Channel

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Abstract

Quadrature Phase Shift Keying (QPSK) is a digital modulation technique which transmits two bits per symbol without additional bandwidth. This project implements a complete, real-time QPSK transmitter and receiver entirely on a Spartan-7 FPGA (Boolean Board, XC7S50). The system performs bit-to-symbol mapping, carrier generation, quadrature mixing, and matched low-pass filtering, and includes a simulated AWGN channel with live-adjustable noise level for testing. System performance is validated through bit-error-rate (BER) measurement and live constellation visualization streamed over UART. This report defines the problem statement, reviews relevant literature on QPSK theory and existing FPGA-based QPSK implementations, and outlines the architecture and verification plan for the mid-term and final phases. Carrier recovery via a Costas loop, symbol timing recovery via the Gardner algorithm, and a coded-versus-uncoded BER comparison are being considered as optional extensions if time permits.

1. Introduction

Modern communication systems rely heavily on digital modulation to move information reliably and efficiently over a band-limited channel. Digital modulation schemes vary properties of a carrier, such as its amplitude, phase, or frequency, to represent information while making efficient use of the available communication bandwidth.

Quadrature Phase Shift Keying (QPSK) is one such scheme. Unlike Binary PSK, which uses two carrier phases to represent one bit per symbol, QPSK uses four equally spaced carrier phases, so that each transmitted symbol represents two bits at once [2]. This is conveniently described using an in-phase (I) and quadrature (Q) component, which together represent the two bits of each transmitted symbol. The transmitted signal can therefore be represented as a combination of cosine and sine carriers at the same frequency, with the I and Q components determining the cor-

responding symbol.

Field-Programmable Gate Arrays (FPGAs) provide re-configurable digital hardware rather than a fixed instruction-execution pipeline: operations such as carrier generation, multiplication, and filtering can each be realized as dedicated, concurrently operating hardware blocks. This matters for a real-time communication system, where all stages of modulation and demodulation must run continuously at the signal's own rate. Prior work has already demonstrated that a complete QPSK modulator and demodulator can be realized on FPGA hardware [1, 3], which motivates implementing our system on FPGA rather than restricting it to a Python/MATLAB simulation alone.

Narrowing from this general motivation to our specific system, this project implements a complete QPSK transmit-receive chain in which a bit stream is first mapped to QPSK symbols, carriers are generated by an NCO, the symbols are modulated onto these carriers, and the resulting signal is passed through a simulated AWGN channel. At the receiver, the signal is coherently demodulated, passed through an FIR filter, and a symbol decision block recovers the original bit stream. Each of these stages is implemented as a distinct hardware block on the Spartan-7 FPGA (Boolean Board, XC7S50).

To validate this system, we follow a staged verification approach: a software reference model is developed first, followed by a fixed-point version of the same algorithm, and finally the FPGA/RTL implementation itself. At each stage, the implementation's behavior is compared against the reference model, and the overall system is evaluated using bit-error-rate (BER), constellation diagrams, behavior under varying noise conditions, and FPGA implementation characteristics such as resource utilization.

2. Literature Review

The most directly relevant prior work is Popescu *et al.* [3], who implement a complete QPSK modulator, channel, and demodulator rather than a modulator alone. Their design uses two Spartan-3E boards, with one board im-

plementing the modulator and the other implementing the demodulator. Sine and cosine carriers are generated internally from a ROM and combined with the odd and even bit sequences to form the QPSK signal. At the receiver, the received signal is multiplied by locally regenerated sine and cosine references, and the recovered signals are compared with a threshold to reconstruct the transmitted bits. This work demonstrates the feasibility of a complete FPGA-based QPSK transmit-receive chain and is closely related to our proposed system. Our implementation differs in that the complete digital processing chain is targeted to a single Spartan-7 FPGA rather than two separate FPGA boards.

Carrier generation is an important hardware block in a digital PSK modulator. Al Safi and Bazuin [1] present FPGA implementations of BPSK and QPSK modulators using accumulator-based addressing and lookup-table techniques. For QPSK, the required phase-shifted sinusoidal waveforms are generated using accumulators and a single lookup table, with the implementation carried out in VHDL. This work is directly relevant to our carrier-generation block, where a phase-accumulator-driven numerically controlled oscillator (NCO) with a lookup table is used to generate the sine and cosine references for modulation and demodulation.

Not every FPGA QPSK design follows the same I/Q-based structure. Pareek [2] compares a conventional I/Q approach with a digitized-carrier approach. In the proposed digitized approach, the input bit pair determines the required phase, and a 4:1 multiplexer selects the corresponding phase of the digitized carrier. Both architectures are implemented on a Spartan-3E FPGA using Verilog HDL. The work therefore demonstrates that QPSK modulation can be implemented using different hardware architectures. Our project adopts the conventional I/Q and carrier-generation approach rather than the multiplexer-based phase-selection approach.

Taken together, these works demonstrate FPGA-based QPSK modulation, complete modulator-channel-demodulator systems, and different approaches to digital carrier generation and hardware architecture. The objective of this project is not to introduce a new QPSK algorithm, but to integrate established techniques into a self-contained and verifiable FPGA-based system. The proposed implementation targets a single Spartan-7 XC7S50 FPGA and includes QPSK mapping, NCO-based carrier generation, modulation, a simulated AWGN channel, coherent demodulation, FIR filtering, symbol decision, and BER measurement. Selected I/Q samples will be streamed through UART for constellation visualization on a host computer.

3. Next Steps

References

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